

# Universal Radiative Corrections for Precision Determinations of $V_{ud}$

Ana Costa Pereira

[Ana.Costa-Pereira@liverpool.ac.uk](mailto:Ana.Costa-Pereira@liverpool.ac.uk)

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- ① EW sector and CKM matrix
- ② Operator product expansion
- ③ Effective Weak Hamiltonian
- ④ Wilson coefficients
- ⑤ NNLO corrections

- SM shortcomings: Origin of flavour, CP problem, dark energy, dark matter, matter/anti-matter asymmetry
- Flavour origins is tied to the Higgs mechanics and Yukawa couplings
- EW sector:  $SU(2) \times U(1)$

$$\mathcal{L}_{\text{EW}} = -\frac{1}{4}W_{\mu\nu}^a W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \sum_{\text{fermions}} \bar{\psi} i\gamma^\mu D_\mu \psi$$

$$+ (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi)$$

$$- \left( \bar{Q}_L Y_d \Phi d_R + \bar{Q}_L Y_u \tilde{\Phi} u_R + \bar{L}_L Y_e \Phi e_R \right)$$

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2$$

$$D_\mu = \partial_\mu - ig \frac{\tau^a}{2} W_\mu^a - ig' \frac{Y}{2} B_\mu$$

$$W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g\epsilon^{abc} W_\mu^b W_\nu^c$$

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

Figure 1: CKM matrix

- Complex constants
- CKM unitary:  $V_{CKM}^\dagger V_{CKM} = 1$

The current experimental values for the CKM, from [?]

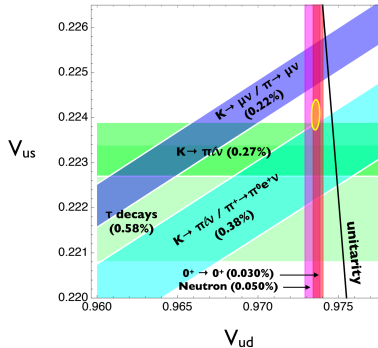
$$\begin{pmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{pmatrix} =$$

$$\begin{pmatrix} 0.97373 \pm 0.00031 & 0.2243 \pm 0.0005 & 0.00394 \pm 0.00036 \\ 0.221 \pm 0.004 & 0.987 \pm 0.011 & 0.0422 \pm 0.0008 \\ 0.0081 \pm 0.0005 & 0.0394 \pm 0.0023 & 0.9991 \pm 0.0001 \end{pmatrix}$$

The CKM matrix is expected to be unitary in the first row,

$$\Delta_{CKM} = 1 - |V_{ud}|^2 - |V_{us}|^2 - O|V_{ub}|^2 = 0 \quad (1)$$

where  $|V_{ub}|^2 \approx 1.5 \times 10^{-5}$

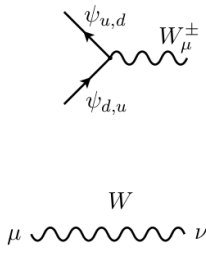


**Figure 2:** Summary of constraints on  $V_{ud}$  and  $V_{us}$  (assuming the Standard Model hypothesis) from nuclear, nucleon, meson, and  $\tau$  lepton decays., [?].

# Weak decays

Neutral interactions - no change of flavour at tree-level;

Charged interactions - mediated by the  $W$  bosons - flavour changing.



$$i \frac{g}{\sqrt{2}} \gamma_\mu \frac{1 - \gamma_5}{2}$$

$$\frac{-ig_{\mu\nu}}{k^2 - M_W^2 + i\epsilon}$$

Figure 3: Feynman rules for charged currents

- NNLO corrections for semi-leptonic decays
- Operator Product Expansion (OPE) and Effective Weak Hamiltonian
- Dim-Reg and Renormalization
- Matching and Wilson Coefficients for pion decay



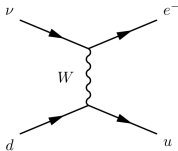


Figure 4: SM tree-level.

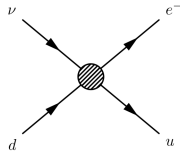


Figure 5: EFT Tree-level.

Evaluating the amplitude  $A_{tree}$ :

$$A_{tree} = \frac{-G_F}{2} V_{ud} \frac{g^{\mu\nu}}{k^2 - m_W^2} [\bar{d}\gamma_\mu P_L u] [\bar{\nu}\gamma_\nu P_L l] \quad (2)$$

but if we consider small internal momenta  $k \ll m_W \approx 80 \text{ GeV}$  we can expand the amplitude. The leading term is given by

$$A = \frac{-G_F}{2} V_{ud} \frac{-1}{m_W^2} [\bar{d}\gamma_\mu P_L u] [\bar{\nu}\gamma_\mu P_L l] = \frac{-G_F}{2} V_{ud} \frac{-1}{m_W^2} [\bar{d}u]_{V-A} [\bar{\nu}l]_{V-A} \quad (2)$$

**OPE:** the product of two charged current operators is expanded into a series of local operators, whose contributions are weighted by effective coupling constants, the Wilson coefficients.

**Effective weak Hamiltonian**

$$H_{eff}(x) = \frac{G_F}{\sqrt{2}} V_{CKM}^i C_i(\mu) Q_i(x) \quad (4)$$

where  $G_F$  is the **Fermi constant**

$V_{CKM}^i$  are the **CKM matrix elements**

$C_i$  are the **Wilson coefficients**  $\longrightarrow$  short distance behavior

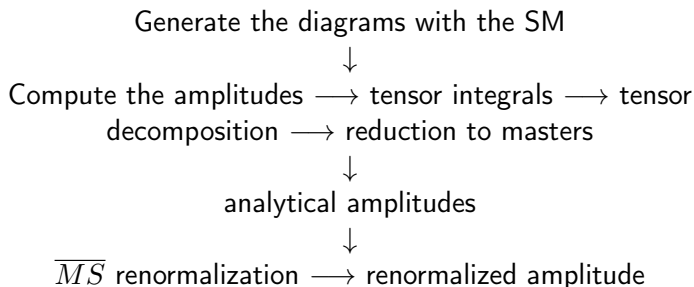
$Q_i$  are the **local operators**  $\longrightarrow$  long distance behavior

$$Q_1 = (\bar{d}\gamma^\mu P_L u)(\bar{\nu}\gamma^\mu P_L e)$$

$$E_1 = (\bar{d}\gamma^\mu\gamma^\nu\gamma^\beta P_L u)(\bar{\nu}\gamma_\mu\gamma_\nu\gamma_\beta P_L e) - (16 - a_q^1\epsilon)Q_1$$

$$E_2 = (\bar{d}\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\gamma^\sigma P_L u)(\bar{\nu}\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta\gamma_\sigma P_L e) - (256 - a_q^2\epsilon)Q_1 + 20E_1$$

- d dimensions;
- scheme dependent;



**Choice of kinematics:** external momenta set to zero and all masses below the  $W$  mass are set to zero. Only masses present in our calculation will be  $m_W$ ,  $m_Z$ ,  $m_H$ ,  $m_t$ .

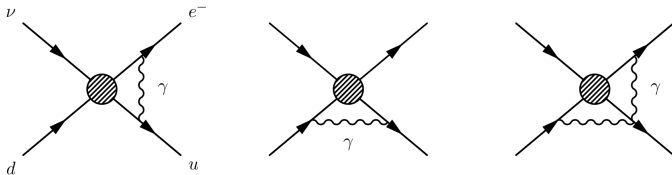


Figure 6: 1 loop EFT contribution for pion decay.

Exact decomposition of a propagator

$$\frac{1}{(k+p)^2 - m^2} = \frac{1}{k^2 - M^2} + \frac{-p^2 + 2k \cdot p - m^2 + M^2}{k^2 - M^2} \frac{1}{(k+p)^2 - m^2} \quad (5)$$

$k$  and  $p$  stand for a linear combination of the internal moment and external momenta. The propagator mass is  $m$  and  $M$  is an introduced fictitious regulator.

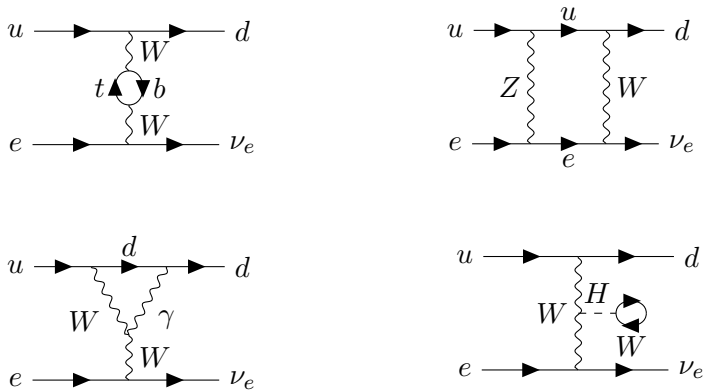
Feynman integral  $\longrightarrow$  regularise the divergences  $\longrightarrow$  introduce parameter that isolates the divergences

**DimReg**  $\longrightarrow$  space-time dimension is extended to  $4 \longrightarrow d = 4 - 2\epsilon$ . The singularities are the poles when  $\epsilon \longrightarrow 0$

$$2\text{-loop}: \frac{a_1}{\epsilon^2} + \frac{b_1}{\epsilon} + c_1 \quad (6)$$

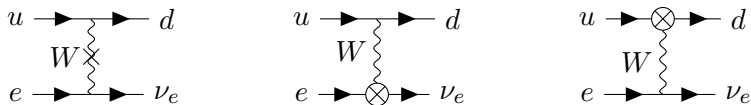
with  $a_1$ ,  $b_1$  and  $c_1$  being constants.

# 1 PI amplitudes in the full SM



**Figure 7:** Some 1 loop diagrams in the full theory for the semi-leptonic decay.

The 1 loop counterterms needed are



**Figure 8:** 1 loop counterterms in the full theory for the semi-leptonic decay.

In the  $\overline{MS}$  scheme we need following renormalization constants

$$Z_{M_W}, Z_{M_Z}, Z_{s_W}, Z_e, Z_W, Z_u, Z_d, Z_{ele}, Z_\mu, Z_\nu$$



SM amplitude proportional to the physical operator  $Q_1$ ,

$$\begin{aligned} M_{SM} = G_F & \left( \frac{\alpha}{4\pi} (a_1 + a_2 \ln \left( \frac{m_t^2}{m_W^2} \right) + a_3 \ln \left( \frac{M_H^2}{m_W^2} \right) + a_4 \ln \left( \frac{M_Z^2}{m_W^2} \right) \right. \\ & \left. + a_5 \ln \left( \frac{\mu^2}{m_W^2} \right) \right) + \frac{\alpha}{4\pi} \frac{(-2)}{\epsilon} \end{aligned} \quad (7)$$

Renormalization of all physical quantities, but still have a divergence: IR divergence.

The Wilson coefficients: full amplitude (proportional to the physical operator  $Q_1$ ) matches the EFT amplitude (proportional to  $Q_1$ ).

$$M_{full} = M_{EFT} \quad (8)$$

$$C_1 = \frac{\alpha}{4\pi} \left( a_1 + a_2 \ln \left( \frac{m_t^2}{m_W^2} \right) + a_3 \ln \left( \frac{M_H^2}{m_W^2} \right) + a_4 \ln \left( \frac{M_Z^2}{m_W^2} \right) + a_5 \ln \left( \frac{\mu^2}{m_W^2} \right) \right) \quad (9)$$

- IR divergences from full theory cancel the UV divergences from the EFT

Normalization to  $G_F$

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}} Q_\mu - \frac{G_F}{\sqrt{2}} V_{ud} C_q Q_q \quad (10)$$

And the Wilson Coefficients are

$$C_r^{LO} = 1 \quad (11)$$

$$C_r^{NLO} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2 \log \left( \frac{\mu^2}{M_Z^2} \right) . \quad (12)$$

- Finite, scheme dependent ( $a_\mu^1$ ,  $a_q^1$ )
- Only box diagrams contribute

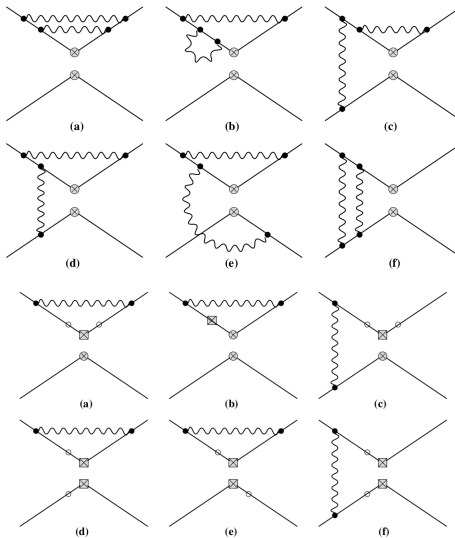


Figure 9: 2 loop EFT amplitudes and CT that contribute to pion decay.

## 2 loop SM contributions

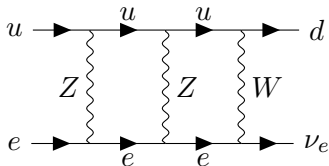


Figure 10: Example of box diagram at NNLO for pion decay.

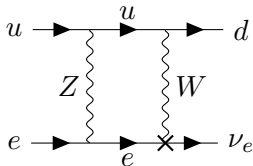


Figure 11: Example of box counterterms at NNLO

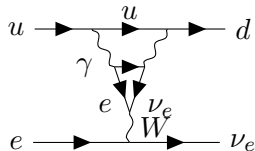


Figure 12: Example of penguin type diagram at NNLO

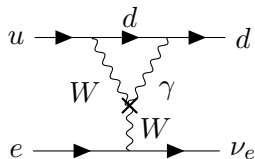


Figure 13: Example of counterterms for penguin diagrams at NNLO

WC depends on  $\mu$ :

$$\mu \frac{d}{d\mu} C_i^f(\mu) = \sum_j \gamma_{ij}^f C_j^f(\mu), \quad (13)$$

depends on active number of flavours;

Anomalous dimension for muon decay (physical operator)

$$\gamma_\mu^{(0)} = 0, \quad \gamma_\mu^{(1)} = \left( \frac{2}{3} N_\ell + \frac{8}{27} n_c N_u + \frac{2}{27} n_c N_d \right) (4 - a_\mu^1), \quad (14)$$

$$\gamma_q^{(0)} = -4, \quad \gamma_q^{(1)} = \left( \frac{16}{9} N_\ell + \frac{64}{81} n_c N_u + \frac{16}{81} n_c N_d \right) (a_q^1 - 8). \quad (15)$$

Solution of RGE for WC

$$C^{(f)}(\mu) = J_f(\mu) U_f(\mu) U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (16)$$

Evolution kernel at LO

$$U_f(\mu) = \left( \frac{\alpha^{(f)}(\mu)}{\alpha(M_Z)} \right)^{-\frac{\gamma^{(0)}}{2\beta^{(0)}}} \quad (17)$$

scheme independence

$$C = U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (18)$$



- Study how to renormalize the SM and EFT amplitudes at NNLO in EW sector ( $\overline{\text{MS}}$  + MOM);
- Computation of the WC at NLO and NNLO for pion and muon decay;
- Checked the contribution of box at 1-loop and box and penguin at 2 loop;